Statistics, Counting & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

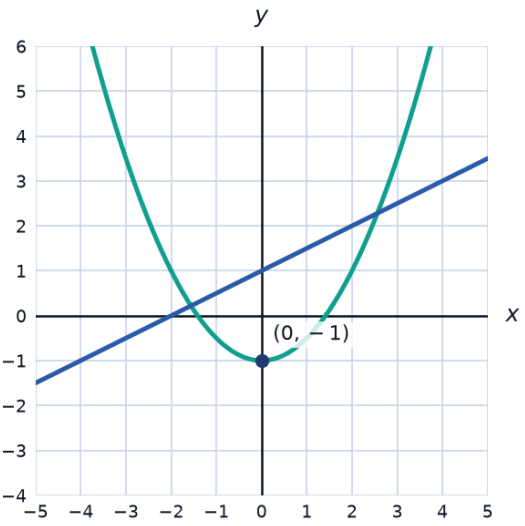
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A box contains 5 red and 3 blue balls. Find the probability that 3 balls drawn without replacement are all red.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $P(A) = 0.6$ and $P(B|A) = 0.4$, find the joint probability $P(A \cap B)$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A box contains 3 red, 4 white, and 5 blue cards. Find the probability of drawing a card that is not white.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the vertical asymptote(s) of $f(x) = \frac{3x}{x^2 - 4}$.

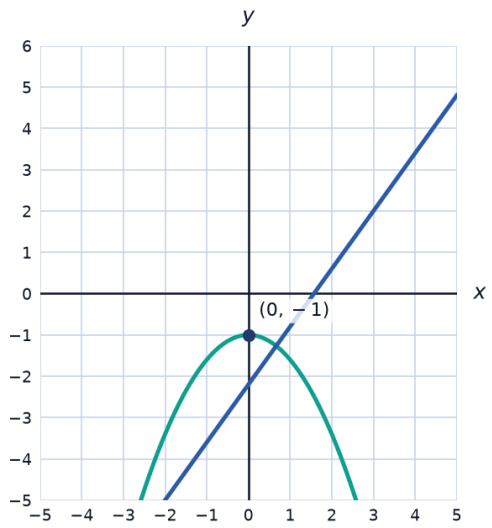
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set has standard deviation 3 and mean 20. If every value in the data set is increased by 5, find the new mean and new standard deviation.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the finite series $\sum_{k=1}^{4} (2k - 1)$.



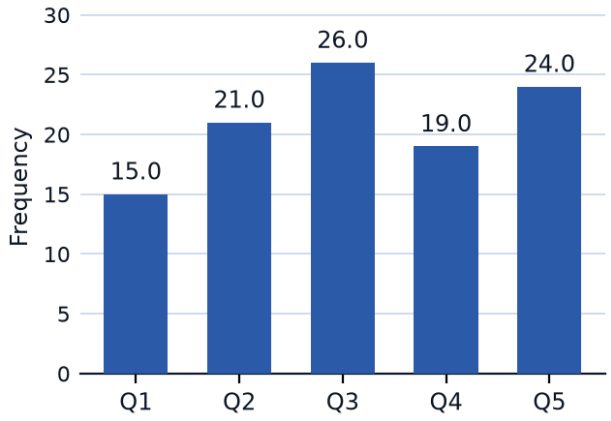
7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Expand $(a + b)^5$ and find the coefficient of the term a^2b^3.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The mean of 10 numbers is 15. One number is removed and the new mean of the remaining 9 numbers is 14. Find the removed number.



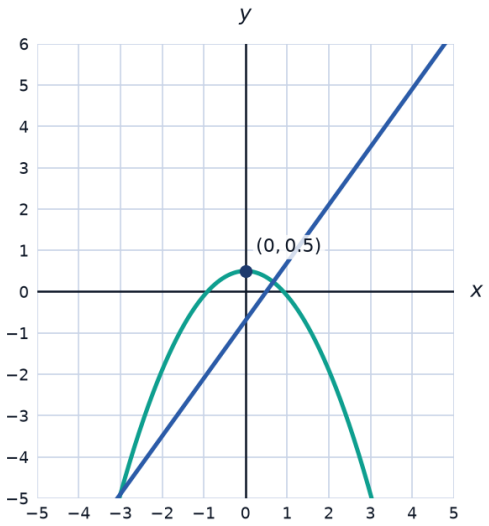
9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the amplitude and period of $y = -4\cos(\frac{\pi x}{3})$.

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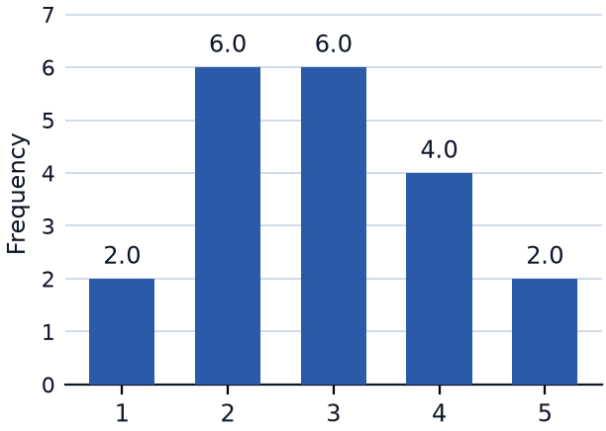
10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the distance from $(1, -2)$ to $(5, 1)$.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

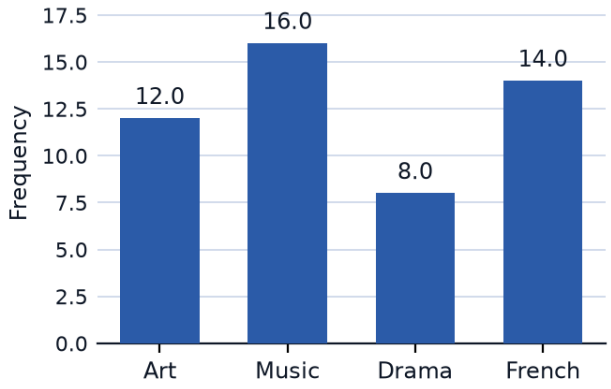
The bar graph below shows the distribution of customer ratings given to a local bakery by 20 customers on a scale from 1 to 5 stars. How many additional 5-star ratings must the bakery receive so that the overall mean rating of all customers becomes exactly 3.5 stars? Show your complete algebraic work.



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12 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The bar graph below shows the number of students enrolled in each of four elective classes at a high school. Each student is enrolled in exactly one of these four classes. If one student is selected at random from these four classes, what is the probability that the student is enrolled in Drama? Show your complete algebraic work.



13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A committee of 4 people is to be selected at random from a group of 10 candidates consisting of 4 mathematicians, 3 physicists, and 3 engineers. Given that the selected committee contains at least one person from each of the three professions, what is the probability that exactly 2 mathematicians are on the committee? Show your complete algebraic work.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Of 31 students, 13 play soccer and 11 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

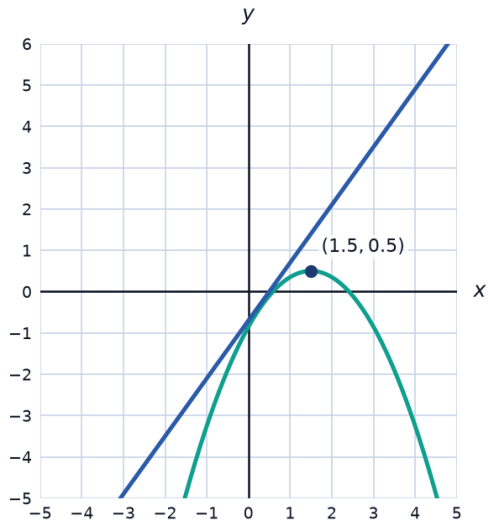
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Advanced Number Theory

A school committee of 4 members is to be selected from a club consisting of 6 juniors and 5 seniors. (a) In how many total ways can the 4 members be selected? (b) In how many ways can the committee be formed if it must contain exactly 2 juniors and 2 seniors?

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Advanced Algebra & Systems

A game pays $x15ifaplayerrollsa6onafairsix-sideddie, pays\6$ if they roll a 5, and the player must pay $x3$ for any other roll (1, 2, 3, or 4). Determine the expected gain or loss per roll, and state whether the game is favorable to the player.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Advanced Functions

A fair coin is tossed 6 times. What is the probability of obtaining exactly 4 heads?

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Trigonometry

In a class of 30 students, 18 play football (F), 14 play basketball (B), and 6 play both sports. How many students play neither sport?

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Pre-Calculus Concepts

The ACT Subject Test in Mathematics consists of 50 questions in 60 minutes. If a student spends an average of 55 seconds on each of the first 30 questions, how many seconds per question are available for the remaining 20 questions?

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

A dataset of 40 exam scores has mean $\bar{x} = 72$ and sample standard deviation $s = 8$. The instructor rescales all scores using $y_i = 1.25x_i - 10$. What are the mean and standard deviation of the rescaled scores?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A charity raffle sells $1,000$ tickets at 5 each. There is one grand prize of $1,000$ and two second prizes of 500 each. What is the expected net gain for a person purchasing one ticket?

a. -3.00

b. -2.00

c. -1.50

d. 0.00

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A random variable X has probability distribution: $P(X = 1) = 0.2, P(X = 2) = 0.5, P(X = 3) = 0.3$. What is the expected value $E(X)$?

a. 2.0

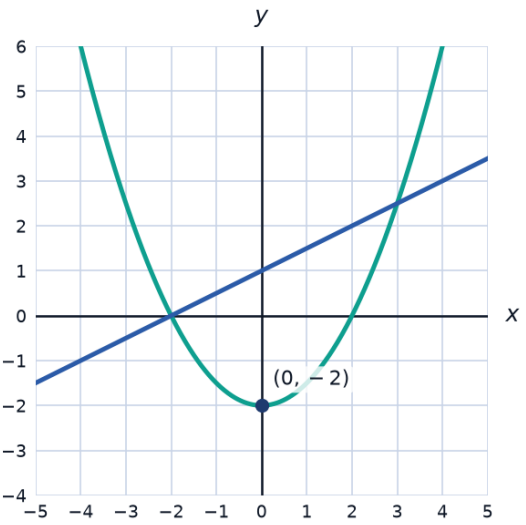
b. 2.1

c. 2.2

d. 2.3

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $E(X) = 4$ and $E(X^2) = 25$, what is the variance $\text{Var}(X)$?



a. 3

b. 5

c. 9

d. 16

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The scores on a standardized exam are normally distributed with a mean of 500 and a standard deviation of 100. Approximately what percentage of test takers score between 400 and 600?

a. 50%

b. 68%

c. 95%

d. 99.7%

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A student scores 84 on a test with mean 72 and standard deviation 8. What is the student's z-score?

a. 1.25

b. 1.50

c. 1.75

d. 2.00

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a normal distribution with mean 50 and standard deviation 5, what value corresponds to a z-score of -1.6?

a. 42

b. 44

c. 46

d. 48

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $P(A) = 0.6, P(B) = 0.5,$ and $P(A \cap B) = 0.3$, what is the conditional probability $P(A \mid B)$?

a. 0.50

b. 0.60

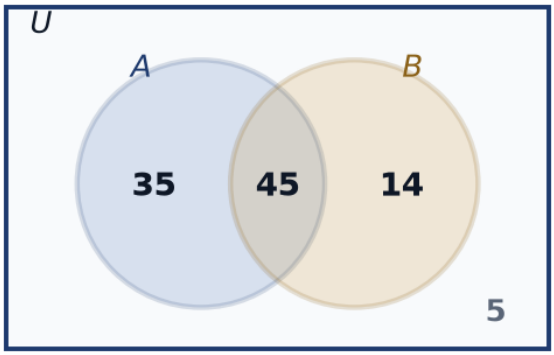
c. 0.75

d. 0.30

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If two events A and B are mutually exclusive with $P(A) = 0.35$ and $P(B) = 0.45$, what is $P(A \cup B)$?



a. 0.70

b. 0.80

c. 0.90

d. 0.1575

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A medical screening test is 95% accurate. If 2% of a population has the disease, what is the probability that a randomly chosen person has the disease AND tests positive?

a. 0.019

b. 0.020

c. 0.095

d. 0.95

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In how many ways can 6 students be seated in a circle?

a. 120

b. 720

c. 60

d. 24

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the minimum number of socks that must be drawn from a drawer containing 10 black socks and 10 white socks to guarantee getting at least one matching pair?

a. 2

b. 3

c. 4

d. 11

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How many distinct permutations can be formed using all the letters of the word 'STATISTICS'?

a. $50,400$

b. $100,800$

c. $302,400$

d. $3,628,800$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A bag has 13 marbles, 2 of which are red. What is the probability of drawing a red marble?

a. $\frac{11}{13}$

b. $\frac{2}{13}$

c. $\frac{1}{13}$

d. $\frac{2}{11}$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

How many ways can you choose 5 items from 13 distinct items, where order does not matter?

a. 1286

b. 1288

c. 154440

d. 1287

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

How many ways can you arrange 2 items chosen from 10 distinct items, where order matters?

a. 89

b. 90

c. 91

d. 45

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A set of exam scores is normally distributed with a mean of $\mu = 74$ and a standard deviation of $\sigma = 8$. What is the z-score corresponding to an exam score of 58?

a. -2

b. -0.5

c. 0.5

d. 2

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The table shows exam marks for 20 students. Find the mean and identify the modal class.

a. Mean ≈ 65, Modal class 60–69

b. Mean ≈ 68, Modal class 70–79

c. Mean ≈ 60, Modal class 50–59

d. Mean ≈ 72, Modal class 60–69

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Trigonometry

A basketball player makes a free throw with probability $2/3$. If she takes 2 free throws, what is the probability she makes at least one?

a. 53/90

b. 55/72

c. 8/9

d. 35/36

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Pre-Calculus Concepts

Given that $\sin\theta = \frac{5}{13}$ and $\frac{\pi}{2} < \theta < \pi$, what is the value of $\cos\theta$?

a. $-\frac{12}{13}$

b. $-\frac{5}{12}$

c. $\frac{12}{13}$

d. $\frac{13}{12}$

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

If $f(x) = 2x - 1$ and $g(x) = x^2 + 3$, what is the value of $g(f(2))$?

a. 9

b. 12

c. 13

d. 21

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Advanced Plane Geometry

If $\sin\theta = -\frac{20}{101}$ and θ is in Quadrant III, find $\cos\theta$.

a. $-\frac{99}{101}$

b. $-\frac{20}{101}$

c. $\frac{20}{101}$

d. $\frac{99}{101}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Advanced Number Theory

If $\cos\theta = \frac{99}{101}$ and θ is in Quadrant IV, find $\sin\theta$.

a. $\frac{20}{101}$

b. $\frac{99}{101}$

c. $-\frac{20}{101}$

d. $-\frac{99}{101}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Advanced Algebra & Systems

If $\cos\theta = -\frac{40}{41}$ and θ is in Quadrant II, find $\sin\theta$.

a. $\frac{9}{41}$

b. $\frac{40}{41}$

c. $-\frac{9}{41}$

d. $-\frac{40}{41}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Advanced Functions

Find the vector projection of $\mathbf{u} = \langle 2, 2\rangle$ onto $\mathbf{v} = \langle 3, 4\rangle$.

a. $\frac{14}{25}\langle 3, 4\rangle$

b. $\frac{14}{5}\langle 3, 4\rangle$

c. $\frac{14}{25}\langle 2, 2\rangle$

d. $\frac{19}{25}\langle 3, 4\rangle$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Advanced Trigonometry

If $\sin\theta = \frac{1}{3}$ in Quadrant I, find the exact value of $\sin(2\theta)$.

a. $7/9$

b. $4\sqrt{2}/9$

c. $-7/9$

d. $-4\sqrt{2}/9$

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